SMART INDIA HACKATHON 2026

Problem Statement ID	**SIH26102**
Problem Statement Title	**Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**
Organisation	**Ministry of Statistics & Programme Implementation (MoSPI)**
Theme	**Smart Automation**
PS Category	**Software**
Team ID	**121911**
Team Name	**NeuralNexus**

SMART INDIA HACKATHON 2026

DRISHTI

The AI-eye on every MPLADS rupee

Detect anomalies. Explain risks. Enable smarter oversight of public funds.

NeuralNexus

DRISHTI

SMART INDIA HACKATHON 2026

PROPOSED SOLUTION

DRISHTI is an AI-powered risk-intelligence layer on top of **eSAKSHI**. **12 AI agents** check every MPLADS work at every stage — before sanction, during construction, on money & time, and after completion — reading not just the records but the **certificates, bills and geo-tagged photos** already on eSAKSHI, and turn what they find into a **daily inspection plan** for District Authorities.

DETAILED EXPLANATION

- **12 specialist agents:** each checks one kind of problem CAG audits have found — cost, duplicates, delays, fake evidence.
- **Reads the evidence:** completion certificates, bills, Hindi utilisation certificates and geo-tagged photos.
- **Explainable Risk Score (0–100):** fuses all agents + top-3 reasons in plain Hindi/English.
- **Inspection Planner + Field App:** picks works for mandatory inspections, routed; verdicts retrain the agents.

HOW IT ADDRESSES THE PROBLEM

- **Manual monitoring can't scale →** every work screened automatically, every day.
- **Money moves before checks →** pre-sanction check inside the 45-day window.
- **Evidence uploaded, never checked →** every photo & bill read, compared and placed on the map.
- **Many alerts, few officers →** risk-ranked queue + optimised visit route.

INNOVATION & UNIQUENESS

- **Reads eSAKSHI's own evidence:** spots reused photos, wrong locations, missing plaques and bills that don't add up.
- **Proven, then measured:** methods India already runs at scale (PMAY-G, Amrit Sarovar); each agent is measured on real data first.
- **Clause-cited, human-in-the-loop:** every flag cites the guideline; flags for review, never accuses.
- **Learns from the field:** officer verdicts become labels, solving the “no fraud labels” problem.

12 AI AGENTS ACROSS THE LIFE OF A WORK — EVERY PS REQUIREMENT COVERED

BEFORE SANCTION

Eligibility — works the rules don't allow
Duplicate & Split — funded twice, or split up
Cost — cost overruns & padded estimates

DURING CONSTRUCTION

Site Progress — claim vs photo, plaque check
Photo Forensics — reused or edited photos
Geo-Location — photo taken elsewhere
Document — bills that don't add up
Satellite — ghost ponds & long roads

MONEY & TIME

Delay Prediction — likely to miss 12 months
Fund Flow — idle funds, year-end rush
Contractor Network — a few firms win all

AFTER COMPLETION

Citizen Verification — asset exists & is used

Then, for every work:
Risk Fusion · Inspection Planner · Ask DRISHTI

NeuralNexus

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

SYSTEM ARCHITECTURE & WORKFLOW

01 — Data Ingestion
- 2.1 lakh eSAKSHI works
- Certificates, bills & photos
- Field app & citizen uploads

→

02 — Data Validation
- Date & amount checks
- Read documents & photo stamps
- Mask names & phone numbers

→

03 — 12 AI Agents
- Before sanction (3)
- During construction (5)
- Money & time (3), after (1)

→

04 — Risk Fusion
- Weighted fusion → 0–100
- Top-3 reasons + evidence
- Low / Med / High / Critical

→

05 — Alerts & Dashboards
- 4 role-based views
- Inspection plan + field app
- SMS / email / PDF brief

12 AI AGENTS — HOW EACH ONE WORKS

Eligibility
How: guideline rules + AI that reads the clause
Catches: works the guidelines don't allow

Duplicate & Split
How: DIGIPIN map grid + Hindi & English text match
Catches: same work funded twice, or split up

Cost
How: cost range of similar works in the district
Catches: estimates far above similar works

Site Progress
How: vision AI checks each claim against photos
Catches: “roof done” with no roof; no plaque

Photo Forensics
How: image fingerprints across all works
Catches: same photo or certificate reused

Geo-Location
How: photo GPS & stamp vs district map
Catches: photos taken elsewhere, or too early

Document
How: AI reads certificates, bills & UCs
Catches: amounts & dates that don't add up

Satellite
How: Sentinel-2 before/after, big works only
Catches: ponds & long roads never built

Delay Prediction
How: model trained on real completion history
Catches: works likely to miss 12 months

Fund Flow
How: MP & district money funnel over time
Catches: idle funds, year-end rush

Contractor Network
How: contractor–agency–district graph
Catches: a few firms winning most works

Citizen Verification
How: QR code at site → photo + GPS
Catches: assets missing or not in use

TECHNOLOGY STACK

Frontend	Next.js (React), Tailwind CSS, MapLibre maps
Backend / APIs	Python FastAPI, role-based access (JWT)
AI / ML	LightGBM, scikit-learn, sentence-transformers
Vision & Documents	Vision AI (Gemini; open-weight option), OpenCV
Maps & Satellite	LGD boundaries, DIGIPIN, Sentinel-2
Data & Security	PostgreSQL + pgvector, hash-chained audit log

NeuralNexus

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

FEASIBILITY ANALYSIS

Technical Feasibility

Proven Methods
Every agent reuses a method India already runs at scale: PMAY-G geo-tagged photos, Amrit Sarovar ponds.

Data Feasibility

Real eSAKSHI Data
2,09,993 works already loaded; 2 in 3 completed works carry certificates or photos to check.

Operational Feasibility

Workflow Fit
Read-only layer on eSAKSHI; fits the MP → District → Agency flow, no new data entry.

Economic Feasibility

Low Cost
Open-source tools; pay-per-use AI at well under ₹1 a document; no licence fees.

CHALLENGES & RISKS

Challenges & Risks	Strategies to Overcome
No Confirmed Fraud Labels Few proven cases to learn from.	**Real Outcomes as Labels** Completion history & planted tests; officer verdicts retrain.
False Alerts Terrain, rates or bulk orders can look like overpricing.	**Independent Checks Must Agree** High risk needs separate agents to agree.
Handwritten & Hinglish Records Hindi certificates and messy text are hard to read.	**Confidence Gate** Unclear readings become “can't tell”, never an alert.
Sensitive Data Named officials, contractors and personal details.	**Fair & Secure by Design** “Needs verification”, never “fraud”; role-based access.

VIABILITY

Human Control
Human-in-the-Loop
AI flags risks; officials verify and decide.

→ **Existing Workflow**
Workflow Compatibility
Works alongside existing eSAKSHI workflows.

→ **Priority-Based Action**
Risk-Based Verification
Picks higher-risk works for mandatory inspections.

→ **Future Expansion**
Modular and Expandable
Agents added one by one; fits PMAY-G & MGNREGA.

PROOF BEFORE LAUNCH

Every agent is tested on real eSAKSHI data and must meet its accuracy target before it goes live. If it fails, it stays switched off.

NeuralNexus

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

IMPACT ON TARGET STAKEHOLDERS

Ministry (MoSPI)

- National risk heat-map by state & district
- Spot systemic issues & guideline gaps
- Evidence-backed reports for Parliament
- Agent accuracy tracked openly

State Nodal Authorities

- District & agency league tables
- Escalate stalled or high-risk works
- Track follow-up on every alert
- Spot contractor concentration

District Authorities

- Daily ranked alert queue with evidence
- Inspection plan with optimised route
- Pre-sanction check in the 45-day window
- One-click PDF evidence brief

Members of Parliament

- Live status of recommended works
- Early warning on delays & idle funds
- Citizens confirm completed assets
- Transparent record builds public trust

BENEFITS

SOCIAL

- Faster delivery of community assets
- Public trust through transparency
- Citizens get assets that truly exist
- Fairer spread of works across areas

ECONOMIC

- Flag risks before funds move
- Targeted visits save officer time
- Catch reused photos & padded bills
- Fewer cost overruns and rework

ENVIRONMENTAL

- Paperless, digital monitoring
- Fewer wasted site trips and fuel
- Durable assets built right first time
- Assets tracked through their life cycle

EXPECTED OUTCOMES

100% of works screened daily; 2,09,993 real works already done

0.13% flagged for review: a queue District Authorities can clear

2 in 3 completed works have certificates or photos DRISHTI can verify

RESEARCH AND REFERENCES

WHAT DRISHTI CHECKS — AND HOW

Capability	How DRISHTI does it	Stage
AI anomaly detection in spending	Cost range of similar works in the same district	**Before sanction**
Duplicate & split works (Hindi + English)	Same-site and bulk-order patterns, meaning match	**Before sanction**
Guideline compliance with para cited	Rules + the clause quoted on every flag	**Before sanction**
Reads certificates & bills (Hindi/English)	Sanction date, payments and contractor read	**During work**
Spots reused or edited photos	Image fingerprints + feature matching	**During work**
Checks where & when photos were taken	GPS stamp read; district, PIN and date checked	**During work**
Delay & stuck-work early warning	Stuck works + each office's track record	**Money & time**
Citizen confirmation & field learning	QR at site; officer verdicts retrain agents	**After completion**
Explainable 0–100 risk score	All agents fused + top-3 reasons	**Every work**
Risk-based inspection route plan	Picks the mandatory 10% of visits, routed	**Every work**

KEY RESEARCH INSIGHTS

₹5 Cr
per MP per year, non-lapsable funds (MoSPI)

~₹4,000 Cr
annual outlay, run on eSAKSHI since Apr 2023 (PIB)

4,918
works without the mandatory MPLADS plaque (CAG audit, 31 districts)

₹280 Cr
in 12,006 works left incomplete (CAG audit, 71 districts)

REFERENCES

Government & Policy Sources

MPLADS Guidelines 2016 & 2023 — mplads.gov.in
eSAKSHI portal & revised fund flow — MoSPI / PIB
CAG Performance Audit of MPLADS — Report 31 of 2010-11
PMAY-G AwaasApp · Mission Amrit Sarovar — MoRD / BISAG-N

Research Papers

Perceptual image hashing — Zauner, 2010
ORB feature matching — Rublee et al., ICCV 2011
Multilingual sentence embeddings — Reimers & Gurevych, EMNLP 2020
LightGBM — Ke et al., NeurIPS 2017

Datasets & Tools

eSAKSHI public works data + uploaded certificates & photos
LGD boundaries · India Post DIGIPIN — official maps
Copernicus Sentinel-2 — free satellite imagery
Gemini vision · OpenCV · scikit-learn